AEMR Network Reliability Analysis

AEMR Case Study | Summary and Objectives | Outage Type Analysis | Outage Type vs. Energy Loss Impact | Top Contributors to Forced Outages | Risk Classification | Outage Duration

AEMR Network Reliability Analysis

AEMR

AEMR is a regulatory body primarily concerned with overseeing the reliability of the American energy grid and ensuring that there are minimal disruptions to its system. Part of this role is examining instances of unplanned, non-consequential (Forced) outages, identifying where they arise from and their contributed loss to the energy systems.

AEMR receives outage reports from its providers, and over a two-year period – from 2016 to 2017 – they have noted a marked rise in the number of reported outages they have received. As a result, the organization would like to take a closer look at this information and address two key areas of concern:

Energy Stability and Market Outages
Energy Losses and Market Reliability

As outages (particularly those that are unplanned), damage the reliability and impact the stability of their markets – AEMR would like to take a regulatory analysis of its providers and outage history across markets in order to better understand these occurrences and identify energy providers that may be putting the whole system at risk.

AEMR Network Reliability Analysis

Number of Outages per Outage Type

Outage Reason	
Forced	2,886
Scheduled (Planned)	700
Consequential	308
Opportunistic Maintenan..	208
Grand Total	4,102

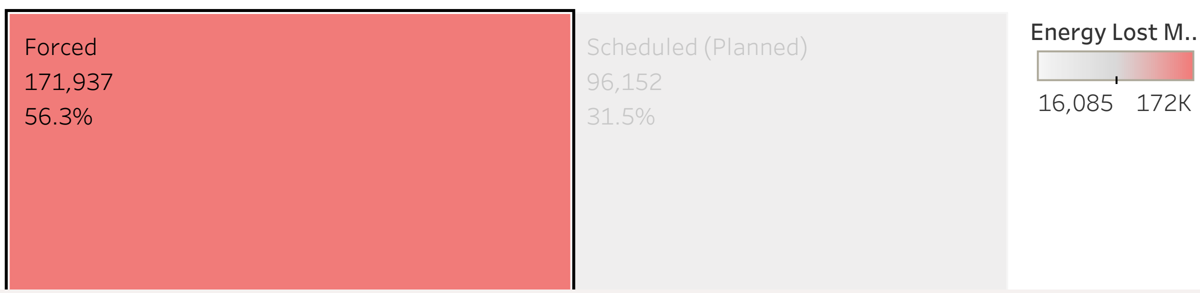
Row_Color
- Other
- Top 1

AEMR Network Reliability Analysis

Which Outage types Have the Largest Impact on Energy Loss?

We have already noted that for AEMR, Forced Outages are of the greatest concern with regards to reliability. Unlike Opportunistic and Planned Outages, which are scheduled to support maintenance, or Consequential Outages, which are caused by external factors (such as weather events, damage to power lines, etc.) — Forced Outages represent a category of precentable outages, that represent a loss of energy and reduced reliability for the energy grid.

Of the 4,102 outages approved by the AEMR, 2,886 can be considered Forced Outages. Based on this value, and based on AEMR's concerns — it would be best to further investigate this category of outages as it pertains to AEMR's main objective.



AEMR Network Reliability Analysis

Summary and Objectives | Outage Type Analysis | Outage Type vs. Energy Loss Impact | Top Contributors to Forced Outages | Risk Classification | Outage Duration | Year over Year Change

Which Providers have the Highest Rate of Forced Outages?

Participant Code	Outage Reason Forced
AURICON	698
GW	544
MELK	334
AUXC	326
TRMOS	237
PUG	159
PJRH	153
KORL	129
PMC	109
COLLGAR	74
STHRNCRS	31
ENRG	28
MUND	19
EUCT	14

Approved Outages

2.0 874.0

We see the highest instance of Forced Outages with:

AURICON

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Risk Classification

(By average Outage Duration in Days)

	Year (Discrete)	
Participant ..	2016	2017
WGUTD	High Risk 1.4	High Risk 6.3
ENRG	High Risk 3.1	High Risk 4.0
MELK	High Risk 2.3	High Risk 3.8
KORL	High Risk 1.9	High Risk 2.0
MUND	High Risk 2.8	Medium Risk 0.5
PJRH	High Risk 1.8	High Risk 1.5
COLLGAR	High Risk 1.5	High Risk 1.7
EUCT	High Risk 2.0	Low Risk 0.3
GW	High Risk 1.1	High Risk 1.3

Calc Main_Risk Classificati..

- High Risk
- Low Risk
- Medium Risk

Risk Classification is calculated primarily based on Outage Duration in Days

Low Risk: Outage Duration < 12 hours (0.5 days)

Medium Risk: Between 12 and 24 hours

High Risk: Outage Duration > 24 hours (1 day)

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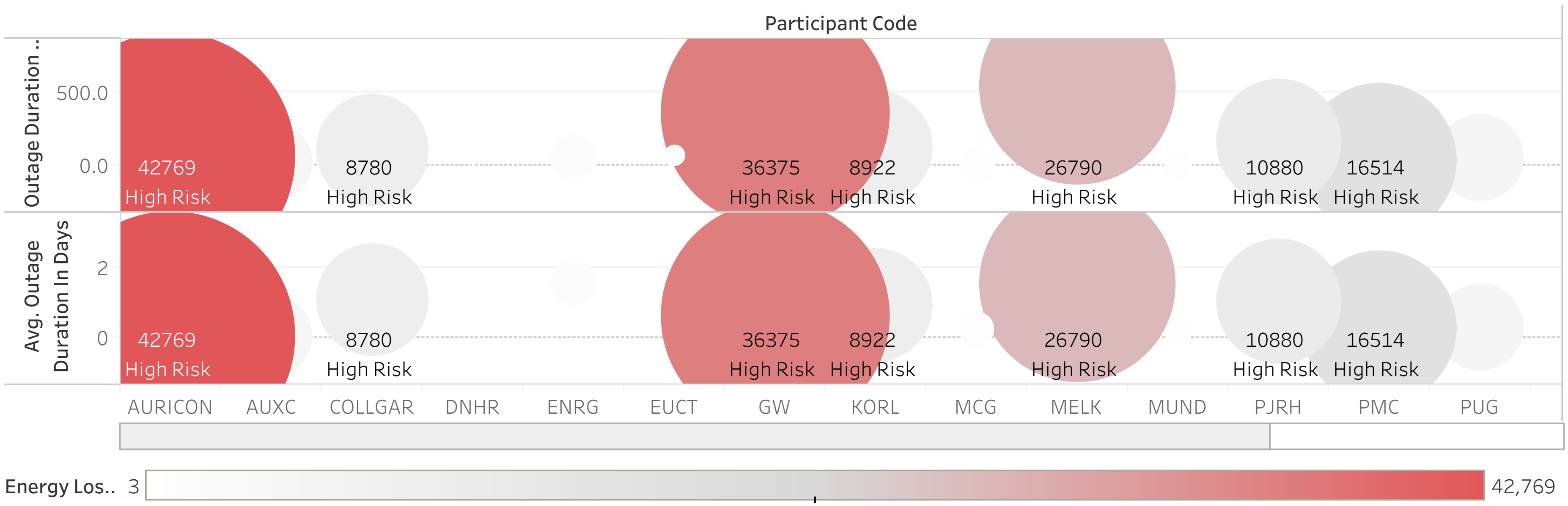
Outage Type Anal..	Outage Type vs. Energy Loss Impact	Top Contributors to Forced Outages	Risk Classification	Outage Duration	Year over Year Change	Conclusion



Participant ..	
AURICON	High Risk
AUXC	High Risk
COLLGAR	High Risk

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Outage Type Anal.. | Outage Type vs. Energy Loss Impact | Top Contributors to Forced Outages | Risk Classification | Outage Duration | Year over Year Change | Conclusion

1 2

Taking a look at year over year changes, we've noted the providers that have seen an increase in their loss statistics over the 2016 to 2017 period. This is particularly helpful in identifying providers where this performance seems consistent, or where loss values may be uncharacteristically high.

Number of Approved Outages

	Year (Discrete)	
Participant Code	2016	201
AURICON		9
	298.0	5
AUXC		-4
	209.0	1
COLLGAR		2
	53.0	
DNHR		
	12.0	
ENRG		-2
	69.0	

Energy Loss MW

	Year (Discrete)	
Participant Code	2016	2017
AURICON		93
	29,737	43,
AUXC		-41
	2,876	1,
COLLGAR		20
	9,647	7,
DNHR		8
	17	
ENRG		-21
	2,110	
EUCT		-39

Outage Duration in Days

	Year (Discrete)	
Participant Code	2016	2017
AURICON		93.6%
	331.7	174.7
AUXC		-41.6%
	48.2	25.4
COLLGAR		20.8%
	103.6	136.3
DNHR		8.3%
	3.5	3.4
ENRG		-21.7%
	250.6	260.5
EUCT		-39.3%
	75.4	5.7

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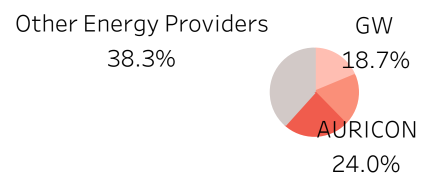
% of Total Energy Loss (in MW)

	Year (Discrete)	
Participant Code	2016	2017
AURICON	19.0%	[illegible]

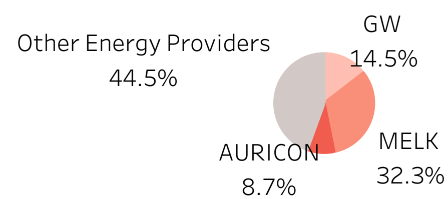
Energy Loss



% of Total Outage Duration in Days

	Year (Discrete)	
Participant Code	2016	2017

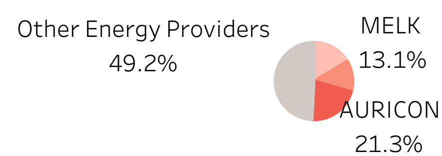
Outage Duration



% of Total Approved Outages

	Year (Discrete)	
Participant Code	2016	2017

Approved Outages



Across the entire dataset, AURICON, GW, and MELK, contributed together – 61 percent of the total energy loss across both years.

MELK, GW, and ENRG contributed 64.3 percent of the total outage duration in days – however AURICON finds itself left out this subcategory by a **slim** margin, landing outside of the top 3 providers by just **.01 percent**.

AURICON, GW, MELK again contribute to about 50.9 percent of the Total Approved outages seen across markets.

We recognize AURICON, GW, and MELK as the loss leaders for AEMR overall. Although AURICON escapes being a top loss leader in the category of outage duration, it notes comparatively high ranking in other categories. It also outranks the providers that have taken its place (ENRG) consistently, as outside of outage duration – this ..